Lecture 6

Chapter 23

Gauss' Law

Gauss' Law

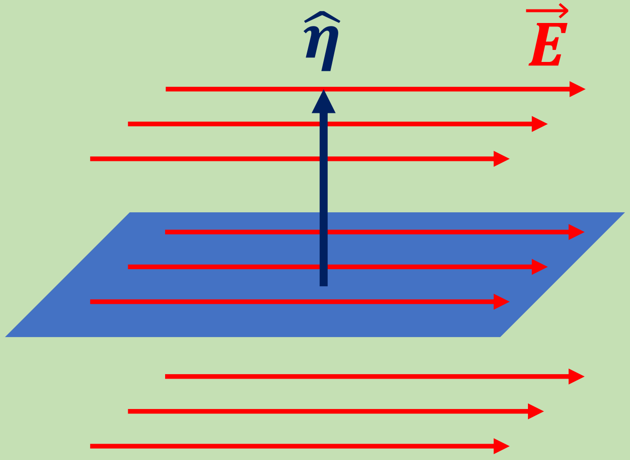
$$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\varepsilon_0}$$

What does it mean?

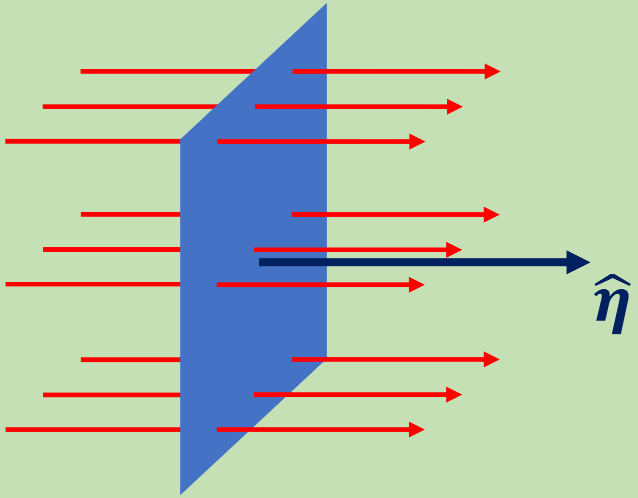
How do we use it?

What does it mean?

- First we need the concept of Electric Flux.



Electric field, $\vec{E}$



Area, A

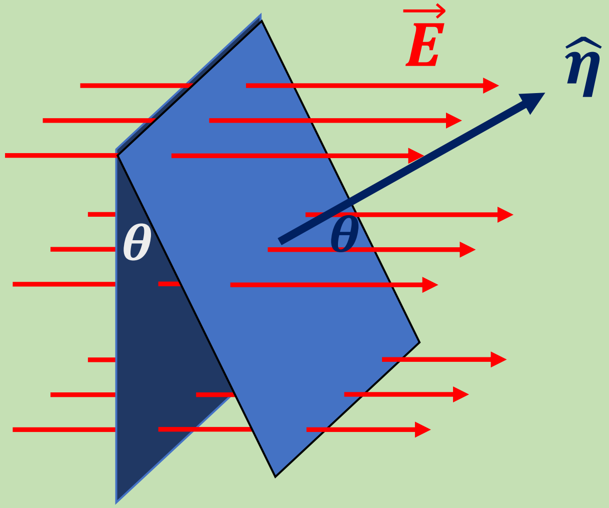
If electric field does not pass through the area, flux is zero.

$$\Phi = 0$$

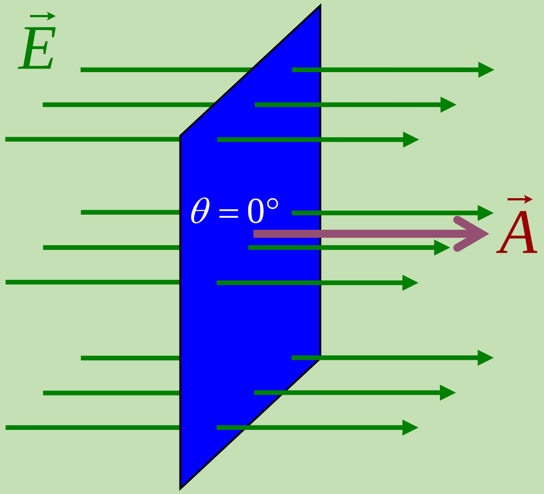
Electric flux is just an electric field perpendicular to the surface times area of that surface, $\Phi = EA$

What does it mean?

- First we need the concept of Electric Flux.

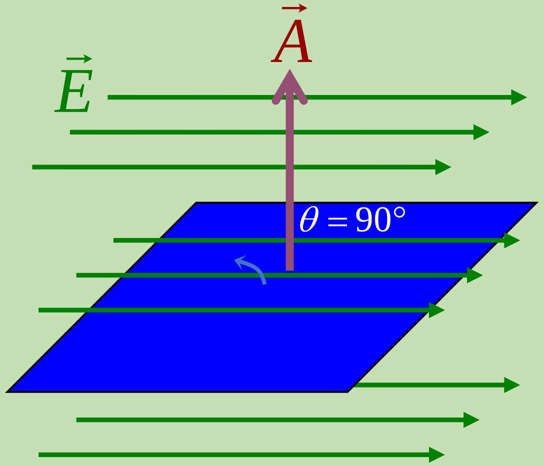


$\cos\theta = 1$



$\Phi = EA$

$\cos\theta = 0$



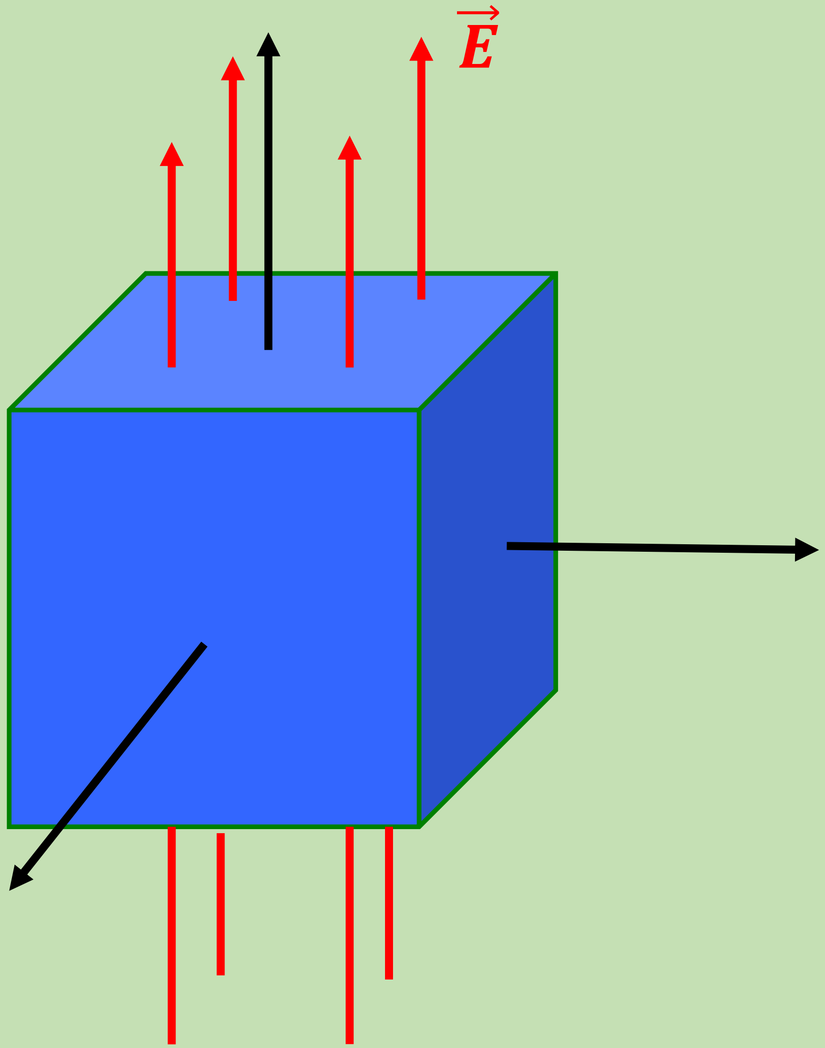
$\Phi = 0$

For more general angles the flux varies as cosθ.

$$\Phi = EA\cos\theta = \vec{E}\cdot\vec{A}$$

What does it mean?

❑**The total flux through a closed surface.**



- We use the convention that the normal always points outward.
- For the four sides,

$$\vec{E}\cdot\vec{A}=0$$

- For the top,

$$\vec{E}\cdot\vec{A}=EA$$

- For the bottom,

$$\vec{E}\cdot\vec{A}=-EA$$

- The total flux is

$$\Phi=EA-EA=0$$

An inward piercing field is negative flux. An outward piercing field is positive flux. A skimming field is zero flux.

$$\Phi = \oint \vec{E} \cdot d\vec{A} \quad \text{(net flux).}$$

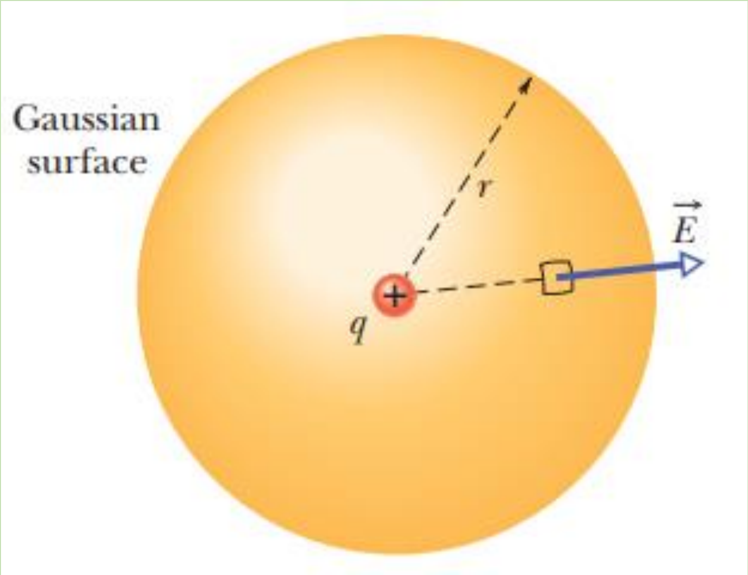
❑Gauss's Law

Gauss' law states that the net flux, $\boldsymbol{\Phi}$ of an electric field, $\vec{\boldsymbol{E}}$ through a closed surface (a Gaussian surface) is equal to the $\frac{1}{\varepsilon_0}$ times of the net charge q_{enc} that is enclosed by that surface.

$$\oint \vec{E}.\overrightarrow{dA} = \frac{q_{enc}}{\varepsilon_0}$$

❑Gauss’ Law and Coulomb’s Law

The electric field, $\vec{E}$, and the unit vector, $\hat{\eta}$ of the smallest surface here in the same direction. So, $\theta = 0^0$



$$\oint \vec{E}.\overrightarrow{dA} = \frac{q_{enc}}{\varepsilon_0}$$

$$\Rightarrow \oint EdA\, cos\theta = \frac{q}{\varepsilon_0}$$

$$\Rightarrow \oint EdA = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E\oint dA = \frac{q}{\varepsilon_0}$$

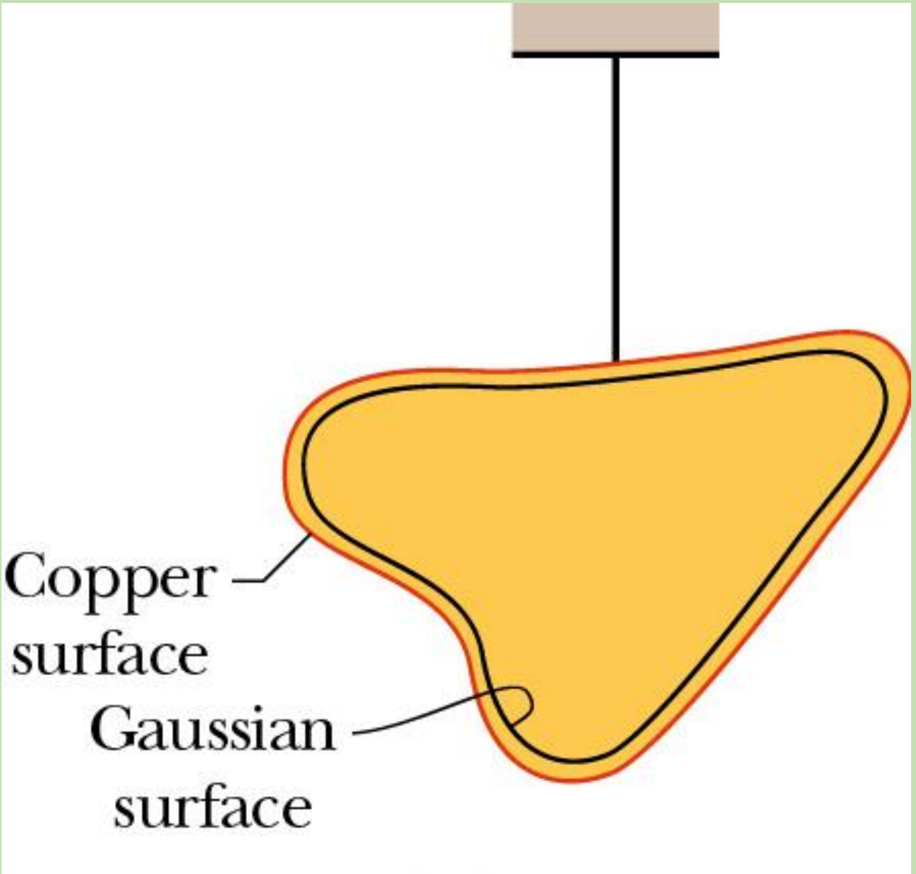
$$\Rightarrow E \times 4\pi r^2 = \frac{q}{\varepsilon_0}$$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

❑A Charged Isolated Conductor

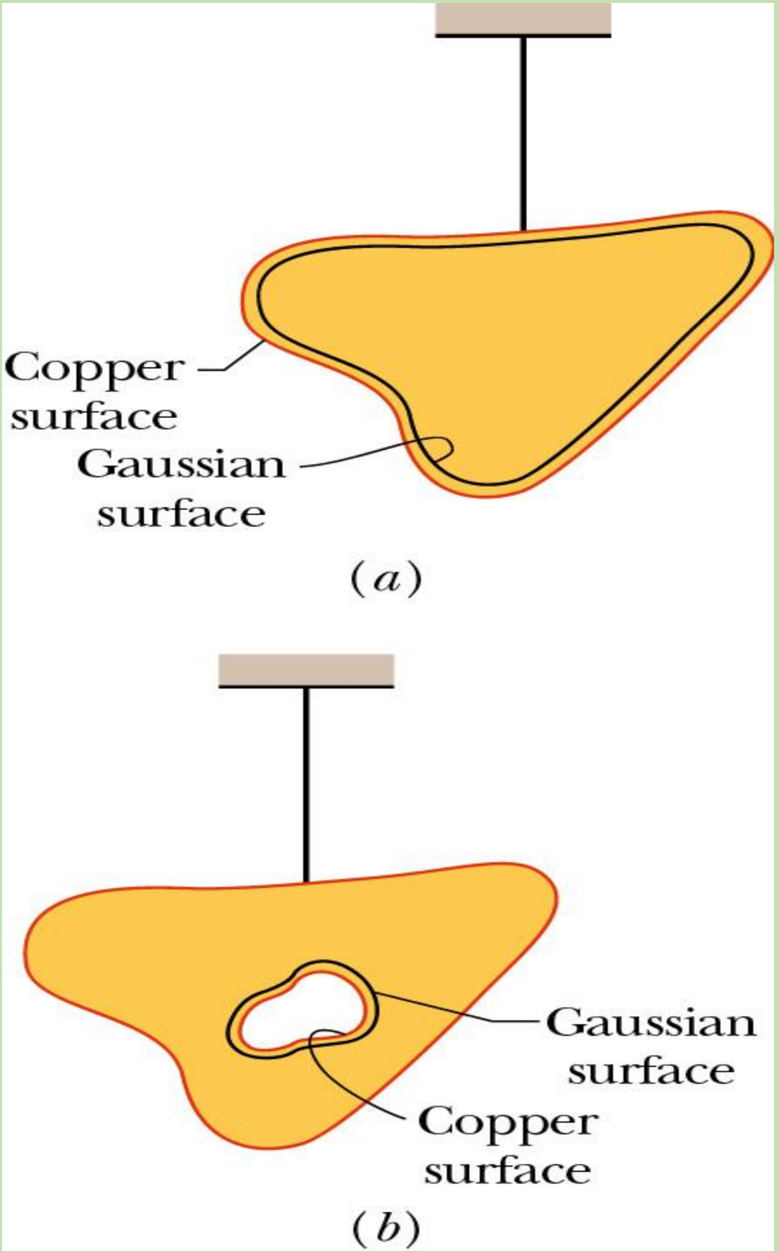
If an excess charge is placed on an isolated conductor, that amount of charge will move entirely to the surface of the conductor. None of the excess charge will be found within the body of the conductor

Field At the Surface of a Conductor



- Imagine an electric field at some arbitrary angle at the surface of a conductor.
- There is a component perpendicular to the surface, so charges will move in this direction until they reach the surface, and then, since they cannot leave the surface, they stop.
- There is also a component parallel to the surface, so there will be forces on charges in this direction.
- Since they are free to move, they will move to nullify any parallel component of E.
- In a very short time, only the perpendicular component is left.

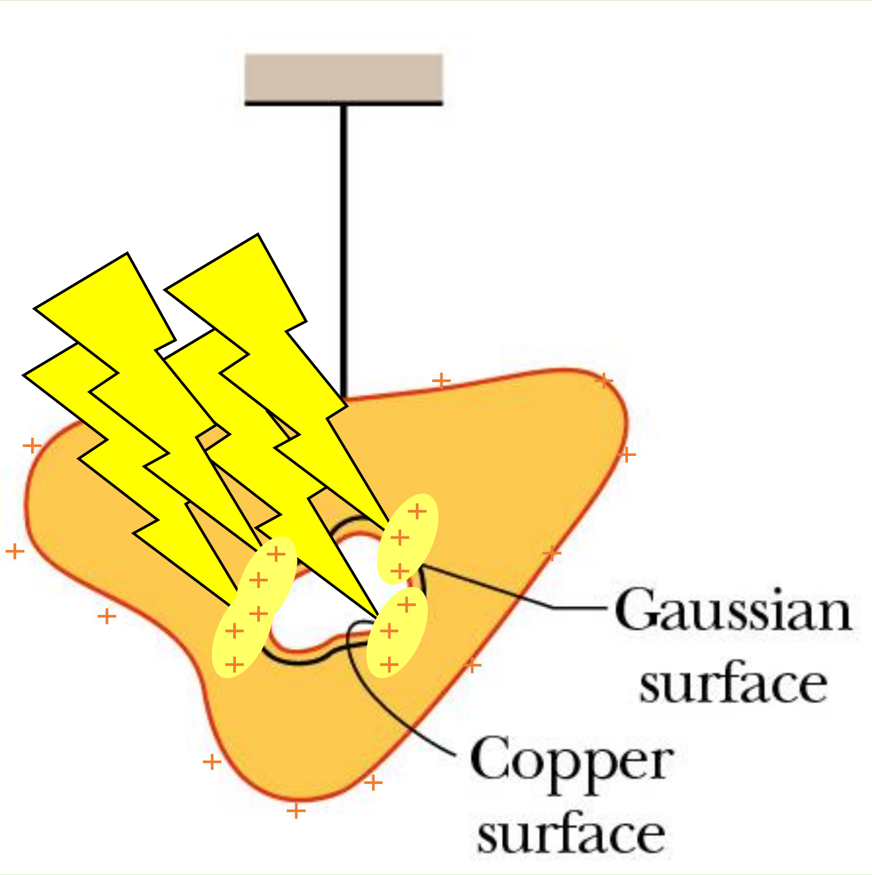
Field Inside a Conductor



We can use Gauss' Law to show that the inside of a conductor must have no net charge.

- Take an arbitrarily shaped conductor having excess charge q, and draw a Gaussian surface just inside, see Fig. (a).
- The electric field inside this conductor must be zero. If this were not so, the field would exert forces on the conduction (free) electrons, which are always present in a conductor, and thus current would always exist within a conductor. (That is, charge would flow from place to place within the conductor.) Of course, there is no such perpetual current in an isolated conductor, and so the internal electric field is zero.
- Since E = 0 everywhere inside, E must be zero also on the Gaussian surface, hence there can be no net charge inside.
- Hence, all of the charge must be on the surface (as discussed in the previous slide).
- If we make a hole in the conductor, and surround the hole with a Gaussian surface, by the same argument there is no E field through this new surface, hence there is no net charge in the hole. [Fig. (b)]

Field Inside a Conductor



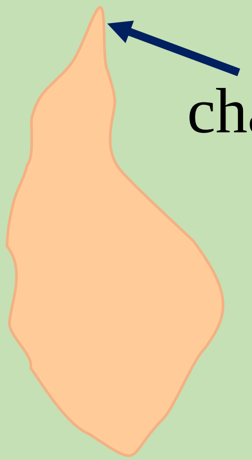
- We have the remarkable fact that if you try to deposit charge on the inside of the conductor...
- The charges all move to the outside and distribute themselves so that the electric field is everywhere normal to the surface.
- This is NOT obvious, but Gauss' Law allows us to show this!

➢ An internal electric field does appear as a conductor is being charged. However, the added charge quickly distributes itself in such a way that the net internal electric field—the vector sum of the electric fields due to all the charges, both inside and outside — is zero. The movement of charge then ceases, because the net force on each charge is zero; the charges are then in electrostatic equilibrium.

There are two ideas here

- Electric field is zero inside conductors
- Because that is true, from Gauss' Law, cavities in conductors have E = 0

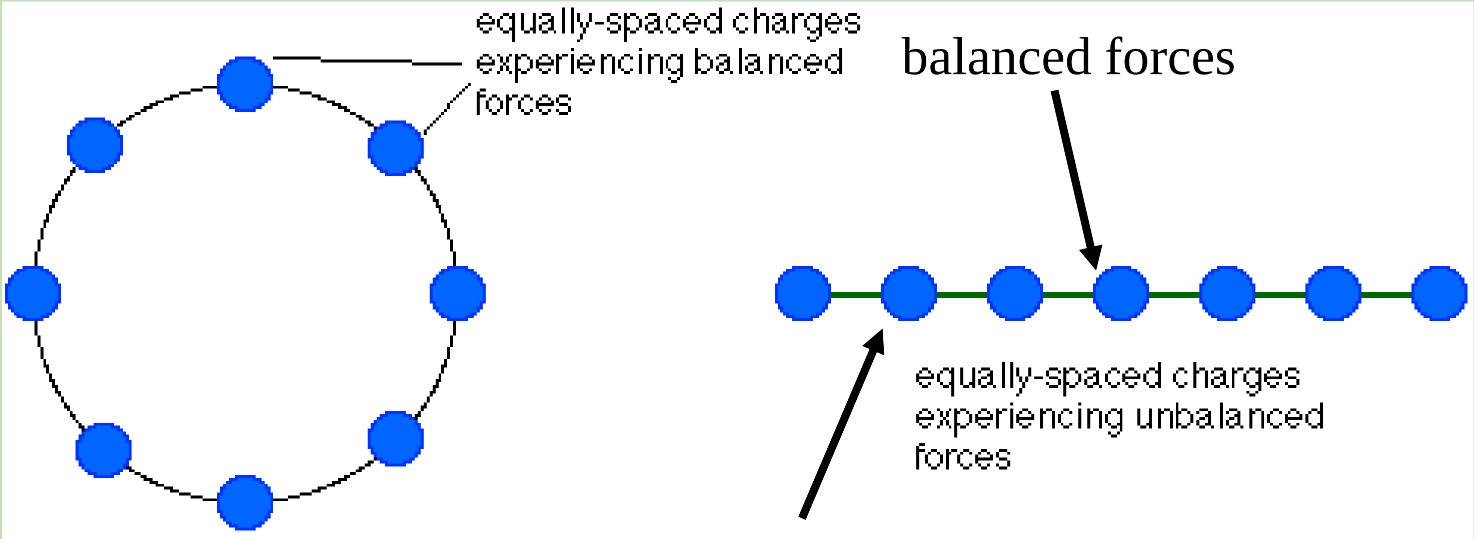
Charge Distribution on Conductors



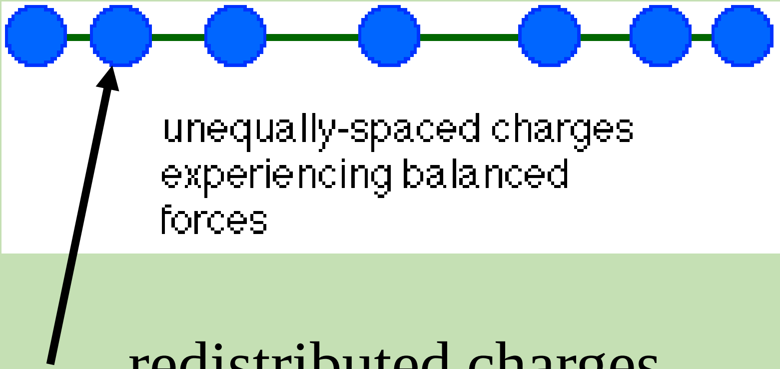
charges are stored near the tip

Conductor

- For a conducting sphere, the charges spread themselves evenly around the surface.
- For other shapes, however, the charges tend to store near sharp curvature.
- To see why, consider a line of charge.



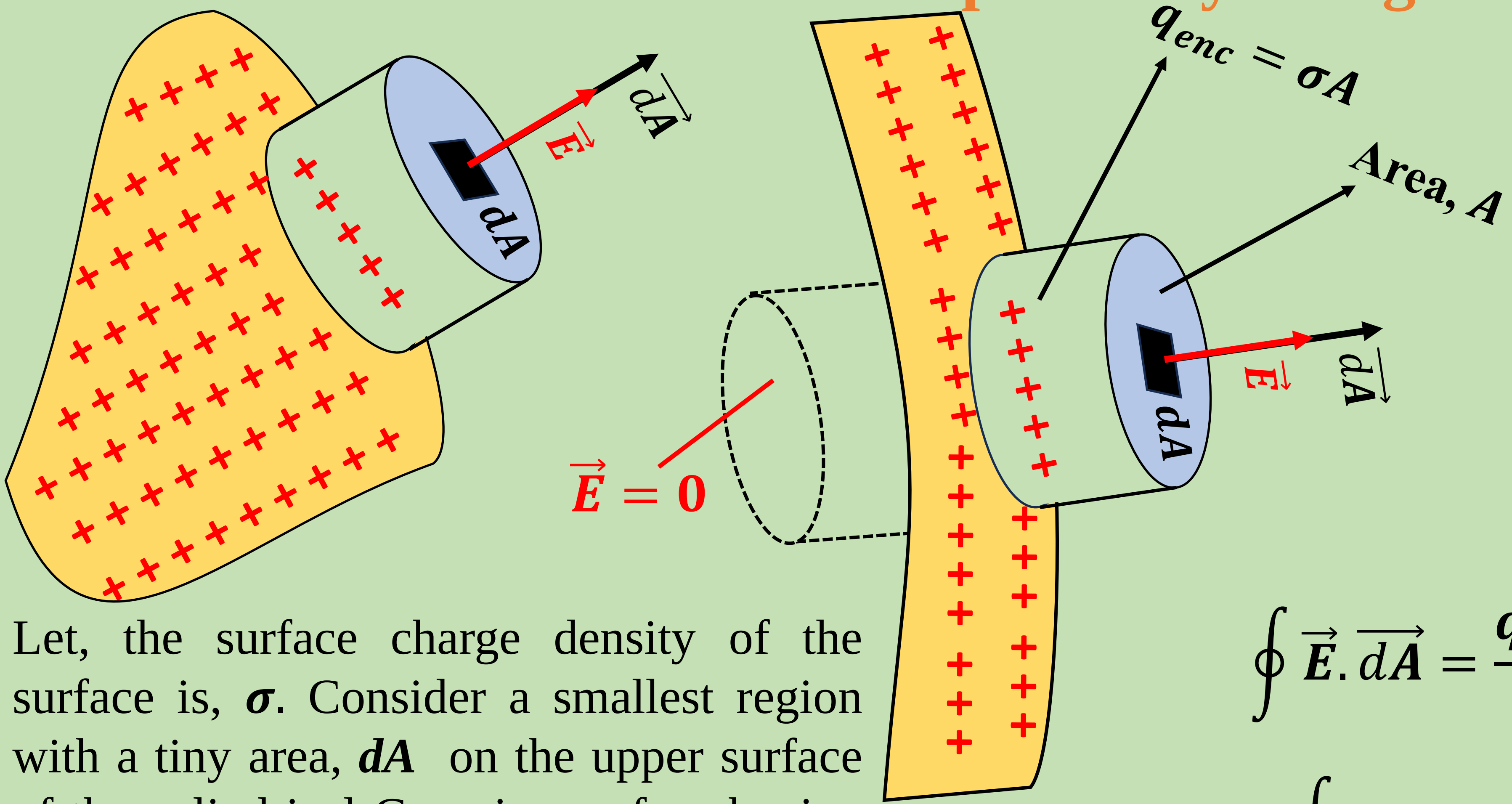
unbalanced forces
(pushed on by one charge from left, but by 5 charges from right)



redistributed charges
(pushed on by one nearby charge from left, but by 5 more distant charges from right)

Some applications of Gauss' Law

❑ The Electric Field Outside the positively charged conductor



Let, the surface charge density of the surface is, $\boldsymbol{\sigma}$. Consider a smallest region with a tiny area, $\boldsymbol{dA}$ on the upper surface of the cylindrical Gaussian surface having an area, $\boldsymbol{A}$ (upper surface) on the flat surface and the Gaussian surface containing charges, $\boldsymbol{q_{enc}}$

$$\oint \vec{\boldsymbol{E}}.\vec{\boldsymbol{dA}} = \frac{\boldsymbol{q_{enc}}}{\boldsymbol{\varepsilon_0}}$$

$$\Rightarrow \oint E dA = \frac{q_{enc}}{\varepsilon_0}$$

$$\Rightarrow E \oint dA$$

$$= \frac{q_{enc}}{\varepsilon_0}$$

$$\Rightarrow EA = \frac{q_{enc}}{\varepsilon_0}$$

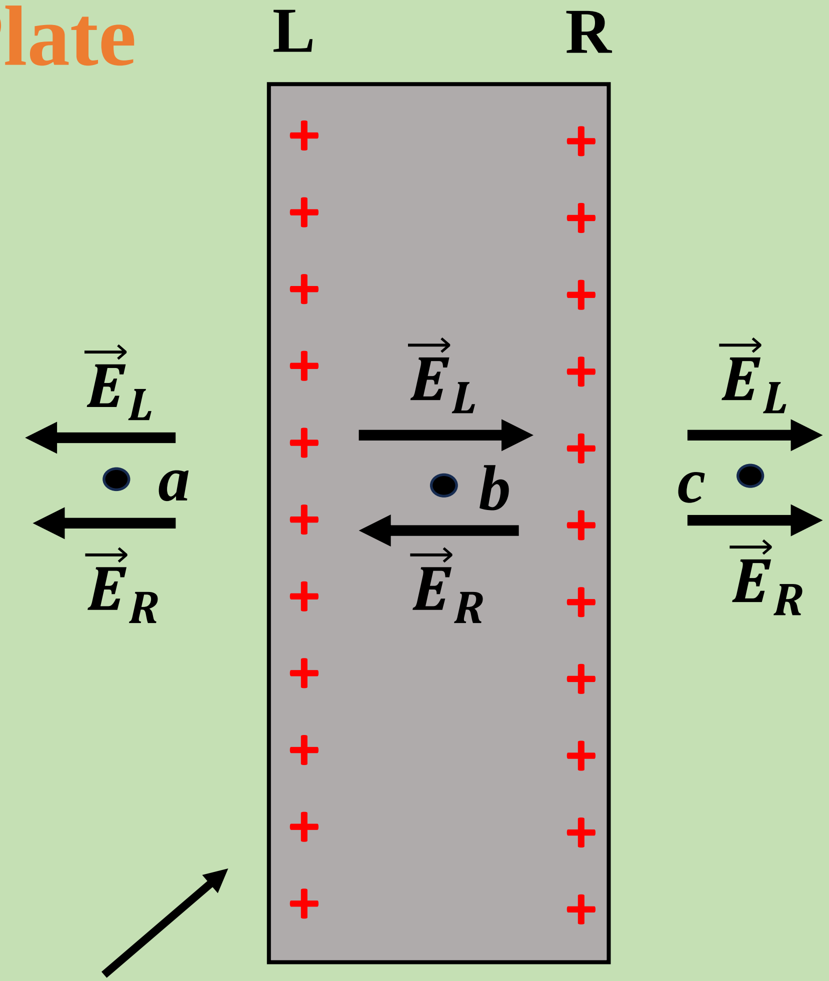
$$\Rightarrow EA = \frac{q_{enc}}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\frac{q_{enc}}{A}}{\varepsilon_0}$$

$$\boldsymbol{E} = \frac{\boldsymbol{\sigma}}{\boldsymbol{\varepsilon_0}}$$

❑ The Electric Field for a Thin Conducting Plate

Suppose, we have a thin conducting plate having an area of $\boldsymbol{A}$. If we spray charge $\boldsymbol{q}$ anywhere on the plate, the charge will distribute itself over both surfaces of the plate, as shown in the figure right side. We therefore expect to find charge $\frac{\boldsymbol{q}}{\mathbf{2}}$ and a charge density $\boldsymbol{\sigma} = \frac{\boldsymbol{q}}{\mathbf{2}\boldsymbol{A}}$ on each surface of the plate.



$$\oint \vec{\boldsymbol{E}}_{\boldsymbol{L}} . \vec{d\boldsymbol{A}} = \frac{\boldsymbol{q_{enc}}}{\boldsymbol{\varepsilon_0}}$$

$$\Rightarrow \oint E_L dA = \frac{q_{enc}}{\varepsilon_0}$$

$$\Rightarrow E_L \oint dA = \frac{q_{enc}}{\varepsilon_0}$$

$$\Rightarrow E_L(A_G + A_G) = \frac{q_{enc}}{\varepsilon_0}$$

Here, A_G is area of Gaussian surface

$$2E_L A_G = \frac{q_{enc}}{\varepsilon_0}$$

$$\boldsymbol{E_L} = \frac{\boldsymbol{\sigma}}{\mathbf{2}\boldsymbol{\varepsilon_0}}; \left(\sigma = \frac{q_{enc}}{A_G}\right)$$

At the same process we can find that

$$\boldsymbol{E_R} = \frac{\boldsymbol{\sigma}}{\mathbf{2}\boldsymbol{\varepsilon_0}}$$

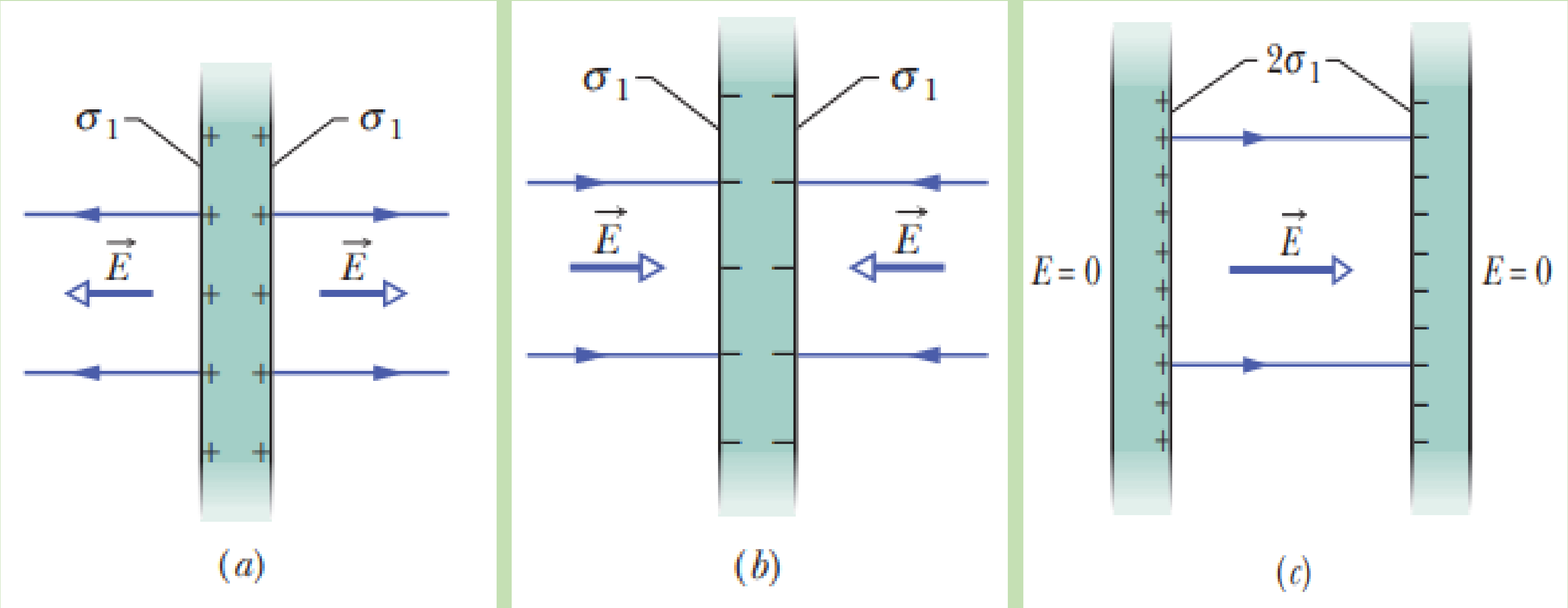
At point $\boldsymbol{a}$, $E = E_R + E_L = \frac{\boldsymbol{\sigma}}{\boldsymbol{\varepsilon_0}}$

At point $\boldsymbol{c}$, $E = E_R + E_L = \frac{\boldsymbol{\sigma}}{\boldsymbol{\varepsilon_0}}$

At point $\boldsymbol{b}$, $E = E_R - E_L = 0$

The electric charge near a thin conducting sheet. Note that surfaces have charges on them. The fields $\vec{\boldsymbol{E}}_{\boldsymbol{L}}$ and $\vec{\boldsymbol{E}}_{\boldsymbol{R}}$ due, respectively, to the charges on the left and right side surfaces reinforce at points A and C, and they cancel at point B n the interior of the sheet.

❑ Two Conducting Plates



(a) A thin, very large conducting plate with excess positive charge.

$$E = \frac{\sigma_1}{\varepsilon_0}$$

(b) An identical plate with excess negative charge.

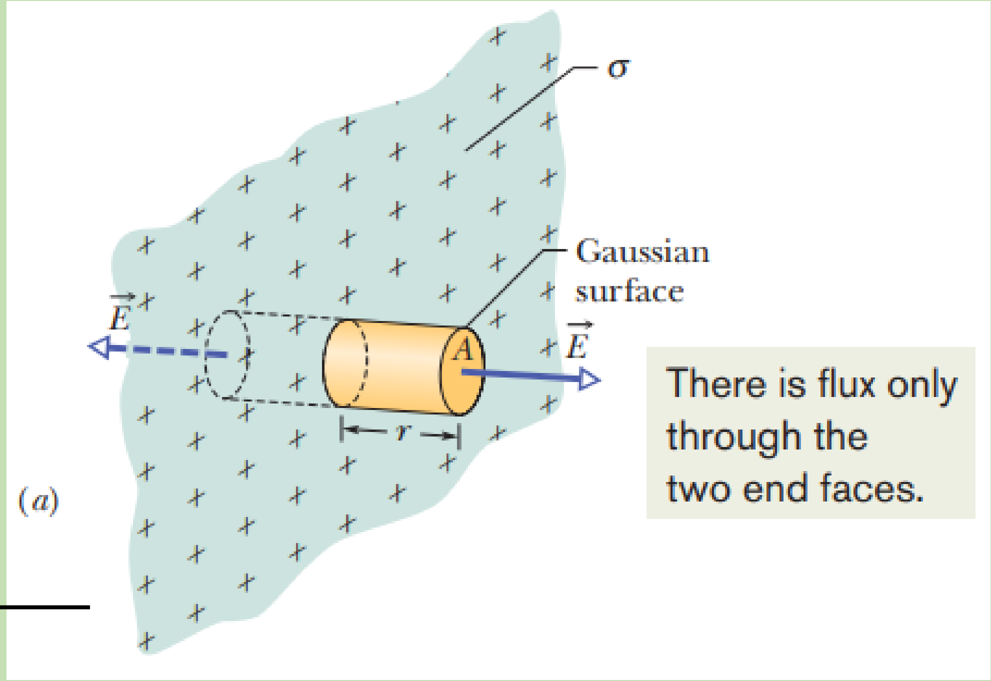
$$E = \frac{\sigma_1}{\varepsilon_0}$$

(c) The two plates arranged so they are parallel and close.

$$E = \frac{2\sigma_1}{\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

❑ Applying Gauss' Law: Planner Symmetry (Nonconducting Sheet)

Let, the surface charge density of the surface is, $\boldsymbol{\sigma}$. Consider a smallest region with a tiny area, $\boldsymbol{dA}$ on the upper or lower surface of the cylindrical Gaussian surface having an area, $\boldsymbol{A}$ (upper or lower surface) on the flat surface and the Gaussian surface containing charges, $\boldsymbol{q_{enc}}$



$$\oint \vec{\boldsymbol{E}}.\overrightarrow{\boldsymbol{dA}} = \frac{\boldsymbol{q_{enc}}}{\boldsymbol{\varepsilon_0}}$$

$$\Rightarrow \oint E dA = \frac{q_{enc}}{\varepsilon_0}$$

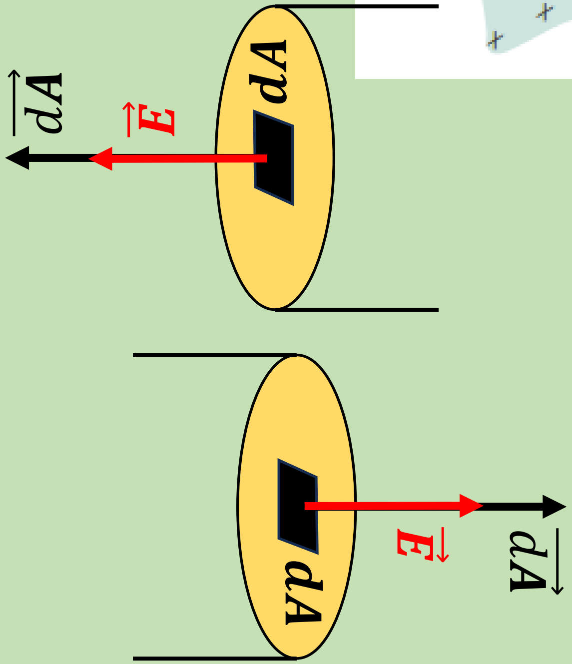
$$\Rightarrow E \oint dA = \frac{q_{enc}}{\varepsilon_0}$$

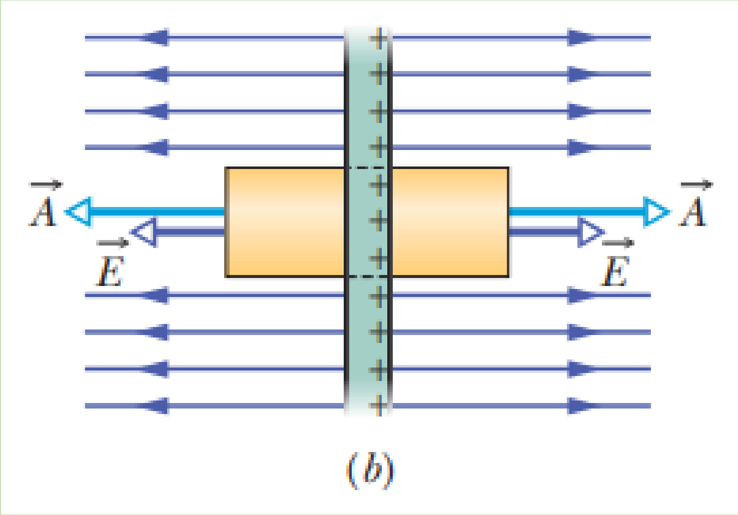
$$\Rightarrow E(A + A) = \frac{q_{enc}}{\varepsilon_0}$$

$$\Rightarrow 2EA = \frac{q_{enc}}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\frac{q_{enc}}{A}}{2\varepsilon_0}$$

$$\boldsymbol{E = \frac{\sigma}{2\varepsilon_0}}$$





Thank You